

AHILYA SANJAY SARNAIK

Dublin, Ireland • +353-858001184 • ahilyacyber@gmail.com • [LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

Detail-oriented and security-focused professional pursuing an **MSc in Cyber Security at the National College of Ireland**, with a strong technical foundation in threat analysis, adaptive defense architectures, secure cloud infrastructure, and federated machine learning systems. Experienced in developing modular intrusion detection systems, building automated AWS security pipelines, and engineering lightweight adaptive honeypots. Adept at secure coding practices, vulnerability assessment, and integrating modern AI security components.

TECHNICAL SKILLS

- **Cybersecurity Domains:** Threat Intelligence, Network Intrusion Detection, Honeypot Architecture, Vulnerability Assessment, SOAR, Application Security, Secure Coding, Prompt Security & AI Firewalls.
- **Cloud & DevOps:** Amazon Web Services (AWS - EC2, IAM, Security Groups), Docker, Containerization, PowerShell, SSH Configuration.
- **Programming & Scripting:** Python, Bash/Shell Scripting.
- **AI, ML & Frameworks:** PyTorch, Flower Framework, Hugging Face Models, Groq API, Autoencoders, Federated Learning (FedAvg, FedProx, SCAFFOLD, FedAdam, Gated Adaptive FL).
- **Tools & Platforms:** Cowrie, Dionaea, Heralding, Elasticpot, NSL-KDD & UNSW-NB15 Datasets, Git/GitHub, Google Colab.

ACADEMIC RESEARCH & PROJECTS

Adaptive Honeypot Architecture for Evasion and Fingerprinting Resistance

MSc Dissertation

- Designed and deployed a lightweight, modular adaptive honeypot architecture aimed at deceiving attackers and mitigating reconnaissance and evasion tactics.
- Orchestrated multi-container deployments using **Docker** running diverse honeypot services including **Cowrie, Dionaea, Heralding, and Elasticpot**.
- Collected, processed, and analyzed latency telemetry and interaction logs to evaluate system resilience against sophisticated attacker fingerprinting techniques.

Federated Learning for Network Intrusion Detection

- Evaluated advanced **Federated Learning** algorithms (FedAvg, FedProx, SCAFFOLD, FedAdam, and Gated Adaptive FL) combined with **Autoencoder** anomaly detection models.
- Utilized **PyTorch** and the **Flower framework** to train models across benchmark datasets (NSL-KDD and UNSW-NB15) while preserving data privacy.

Real-Time AWS Security Orchestration & Automated Response Pipeline

- Built an automated cloud incident response pipeline leveraging **AWS EC2** instances.
- Configured robust remote access mechanisms using **PowerShell** and **SSH**, establishing secure group boundaries and automated event response protocols.

Agentic Firewall Plugin for Large Language Models

- Developed and debugged a secure Python-based **Agentic Firewall plugin** integrated with **Groq API** and **Hugging Face models** in Google Colab.
- Implemented real-time payload filtering and prompt security enforcement rules to protect LLM interactions against malicious injections and bypass attempts.

EDUCATION

Master of Science (MSc) in Cyber Security

Expected Graduation: 2026

National College of Ireland, Dublin | Focus Areas: Network Security, Applied Cryptography, Cloud Security, Threat Modeling, and Secure Software Engineering.

CERTIFICATIONS & CONTINUOUS LEARNING

- **AWS Certified Cloud Practitioner (CLF-C02)**: In Preparation / Coursework Completed.
- **The Cyber Whiskers**: Creator & Author of educational content focusing on secure coding habits, credential isolation, dependency management, and SQL injection prevention.